Periodic Continuous Evaluation Plan — MH OAN API (MahaVistaar)

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Service: `mh-oan-api` (Maharashtra Open Agri Network — Voice Assistant API)

Status: Plan (pre-implementation)

1. What this is

A **weekly, automated quality review** of real farmer conversations.

Every week, a scheduled job pulls a sample of historic chat traces from **Langfuse** (primary telemetry store for this framework), sends each conversation to an **open-source LLM judge** running on **our infrastructure**, and produces:

- Dimension scores (language, accuracy, safety, tools, etc.)
- **Highlighted issues** — e.g. English words left in a Marathi reply, wrong crop name, overly formal tone
- **Suggested corrections** — proposed Marathi phrasing, glossary fixes, prompt gaps (for human review only; nothing auto-deployed)

Results are written back to Langfuse and rolled into a **weekly quality scorecard** for the department and engineering.

This is not ad-hoc log reading. It is a recurring audit loop that turns production traffic into actionable language and quality improvements.

2. Why we are doing this

2.1 The problem

MahaVistaar serves **rural Maharashtra farmers** who depend on clear advice in **Marathi** (and, on the roadmap, **Bhili**). The assistant must use correct agricultural terminology, a conversational tone, and the farmer's selected language — every time.

In practice, language quality issues show up in production even when the underlying agricultural information is correct:

Issue class	What goes wrong	Farmer impact
Language mixing	English crop/scheme/technical terms left in Marathi replies (e.g. "Potassium", "scheme status")	Reduces trust; farmers with low literacy cannot parse the answer
Glossary drift	Terms not taken from the authoritative Marathi glossary; inconsistent translations across sessions	Same concept named differently; confusion across follow-up questions
Register mismatch	Bureaucratic or textbook Marathi instead of farmer-friendly conversational style	Answers feel like government circulars, not a helpful neighbour
Script handling	Roman-script Marathi queries (`gahu` , `kanda`) mapped to wrong Devanagari terms	Wrong crop or input identified; downstream tool searches fail
Language adherence	Response in English when Marathi was selected (or vice versa)	Violates platform policy; breaks voice/TTS pipeline expectations
Transliteration inconsistency	Same English term transliterated differently across replies	Looks unprofessional; erodes confidence in advisories
Bhili gap	Marathi-first responses served to Bhili-speaking tribal farmers without dialect-aware phrasing	Exclusion of tribal communities the state is explicitly targeting

These issues are **hard to catch manually** at scale. The service handles thousands of sessions; reviewers cannot read every trace. Spot checks miss regressions. Prompt or glossary changes can fix one failure mode and introduce another.

2.2 Why periodic LLM review

A **weekly batch LLM judge** over historic traces gives us:

1. **Coverage** — systematic review of K conversations per week, stratified by language and failure signals
2. **Consistency** — same rubric applied every week, comparable scores week-over-week
3. **Discovery** — surfaces patterns humans miss (e.g. "उर्वरक" replaced by English in 12% of fertilizer queries)
4. **Low risk** — suggestions go to humans; no automatic prompt or live-response changes

Research on LLM-as-a-Judge in multilingual and low-resource settings (Doğruöz et al., 2026; arXiv:2607.02235) warns that single judges over-trust their own judgments, especially outside English. This plan addresses that with **multi-judge ensembles**, **target-language rubrics**, and a **monthly human calibration subset**.

3. How this impacts us

3.1 Department (PoCRA / Agriculture)

- **Visibility** — weekly scorecard with mean language scores, issue counts, and exemplar trace IDs
- **Accountability** — evidence that advisories meet language policy before field outreach or scheme campaigns
- **Prioritisation** — "top 10 language issues this week" drives content and glossary review meetings
- **Bhili readiness** — early signal on translation quality as Bhili support rolls out

3.2 Engineering

- **Regression detection** — prompt deploy correlates with drop in `language_quality` score
- **Glossary maintenance** — judge flags terms missing from `glossary_terms.json`; batch updates

- **Trace completeness** — eval job fails loudly if Langfuse traces lack full Q&A or language metadata
- **Tooling feedback** — flags when `search_terms / search_documents` chains produce terminology mismatches

3.3 Farmers (indirect)

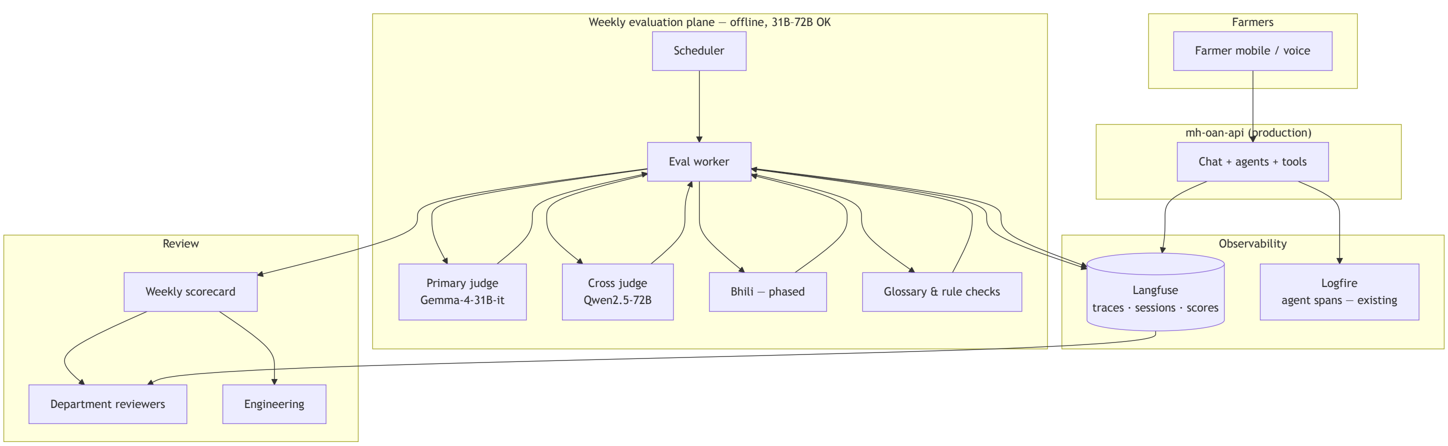
- Better Marathi over time: correct terms, consistent tone, fewer English leaks
- Faster fixes to systematic errors (e.g. wrong transliteration of scheme names)
- Path to **Bhili-native** advisories validated before wide tribal rollout

3.4 What we explicitly do *not* do

- No automatic prompt edits, glossary commits, or live response rewriting
- No farmer PII in judge prompts (Agristack fields remain masked as in production)
- No replacement of human field review for Bhili — LLM judge is a triage layer, not a native-speaker substitute

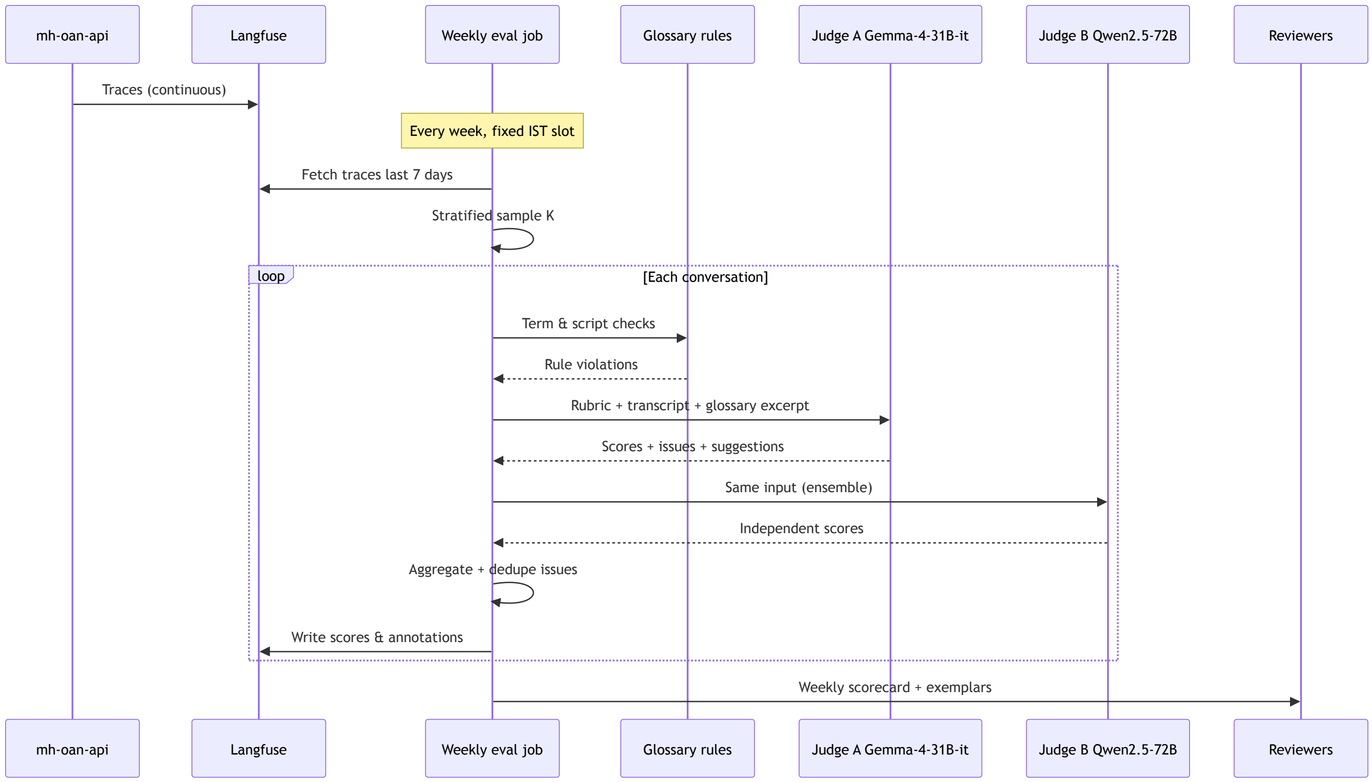
4. System architecture

4.1 Context diagram



Today: Logfire instruments Pydantic AI agents (`instrument=True`). Langfuse export on the chat path is **not yet wired** — first implementation milestone.

4.2 Weekly job sequence



4.3 Component responsibilities

Component	Role
mh-oan-api	Serve farmers; emit complete chat traces
Langfuse	Store historic conversations; receive judge scores and issue notes
Eval worker	Sample, orchestrate judges, aggregate, publish scorecard
OSS judge inference	vLLM on dedicated eval GPUs — models match or exceed the 30B production agent; weekly offline batch, not real-time
Glossary rule layer	Deterministic checks against authoritative term list (English leakage, known term mismatches)
Scorecard	Human-readable weekly summary for triage meetings

5. Weekly process

5.1 Schedule

Parameter	Default
Cadence	Once per week
Window	Previous 7 days of production traffic
Time	Fixed weekday, IST (e.g. Monday 06:00)
Batch size **K**	100–2000 (set from weekly volume and eval GPU capacity)
Judge stack	**Gemma-4-31B-it** primary + **Qwen2.5-72B** cross-judge (see §7)

5.2 Sampling strategy

1. **Stratify** by `target_lang` (Marathi, English; Bhili when live) and top intent categories
2. **Oversample** traces with: errors, empty responses, tool/MCP failures, moderation edge cases
3. **Always include** all safety-flagged or empty-response traces (may exceed K — cap with alert)
4. **Reserve 5%** of K each month for human-labeled calibration (same traces judged by field reviewers)

5.3 Per-conversation judge output

```json
{
  "trace_id": "...",
  "scores": {
    "language_quality": 4,
    "terminology": 3,
    "farmer_friendliness": 5,
    "agricultural_accuracy": 4,
    "language_adherence": 5,
    "tool_use": 4,
    "safety": 5
  },
  "issues": [
    {
      "type": "english_leakage",
      "severity": "medium",
      "evidence": "उत्तरात 'Potassium' इंग्रजीत आहे",
      "suggestion": "ग्लॉसरीनुसार 'पोटॅशियम' वापरा"
    }
  ],
  "suggested_marathi_rewrite": "...",
  "glossary_additions": ["..."],
  "prompt_gap_note": "..."
}
```

5.4 Push-back surfaces

Destination	Content
Langfuse	Per-trace scores, issue tags, reviewer notes
Weekly scorecard	Aggregates, trends, top issue types, exemplar IDs
Department dashboard	Optional roll-up via existing Vistaar telemetry if required

6. Language quality — what we judge

Language quality is the **primary focus** of this framework. Other dimensions (tools, safety, accuracy) support triage but language is the main weekly report section.

6.1 Marathi rubric dimensions

Dimension	Judge question	Failure examples
Language adherence	Is the full response in the selected language?	English paragraph in a Marathi-selected session
Terminology	Are crop, pest, scheme, and input names from standard Marathi agri vocabulary?	"cotton" instead of "कापूस"; wrong pest name
No English leakage	Are technical terms either Marathi or agreed transliteration?	"Fertilizer", "scheme", "warehouse" left in Latin script
Glossary compliance	Do flagged terms match the authoritative glossary?	Glossary says "कीटकनाशक", response says "pesticide"
Register	Is the tone simple, conversational, farmer-friendly?	Formal शासकीय prose for a weather question
Script consistency	Devanagari throughout (except proper nouns per policy)?	Mixed Roman and Devanagari in one reply
Transliteration quality	Are borrowed terms transliterated consistently?	"पोटॅशियम" vs "पोटेशियम" across sessions
Roman input handling	Was Roman Marathi query interpreted correctly?	`gahu` → wrong crop term in response

6.2 English rubric dimensions

Dimension	Judge question
Clarity	Plain language understandable to farmers with basic English
Terminology	Correct agricultural terms; glossary-aligned where bilingual mapping exists

Dimension	Judge question
No inappropriate mixing	No unexplained Marathi fragments in English-selected sessions

6.3 Bhili rubric dimensions (phased)

Bhili (Dehvali dialect, ISO `bhb`) is **extremely low-resource**. Judging strategy is phased:

Phase	When	Approach
P0	Bhili not yet in production	Marathi judge only; monitor Marathi→Bhili translation pipeline separately
P1	Bhili pilot	Marathi judge + AI4Bharat glossary post-edit rules (`mar2bhb` term pairs)
P2	Bhili in production	Dedicated Bhili judge model (fine-tuned or tribal LLM) + monthly native-speaker calibration
P3	Steady state	Ensemble: Bhili judge + Marathi back-translation check

7. Open-source models for evaluation

All judging runs on **our infrastructure** (self-hosted, no vendor API dependency).

Sizing principle: Production runs a **~30B agent**. The weekly evaluator is an **offline batch job** — models **matching or exceeding** production scale are appropriate. Latency is irrelevant; Indic-language quality and human calibration are what matter.

Final model choice requires a **calibration sprint** against 50–100 human-labeled Marathi (and Bhili) conversations before the first department-facing scorecard.

7.1 Marathi — recommended judge stack

Use **Gemma 4** (Google DeepMind, released 2026) as the Indic-language anchor. On IndicParam — 13,207 MCQs across 11 low-resource Indic languages — **Gemma-4-31B-it scores 46.48%**, a **+9.2 point jump** over Gemma 3 27B (37.3%) and competitive with the strongest open models on that benchmark. Gemma 4 also reports **MMMLU 88.4%** and pre-training on **140+ languages** (model card).

Tier	Model	Params	License	Why it fits
Primary	Gemma-4-31B-it	31B dense	Apache 2.0	Latest Gemma; **best open-weight Indic benchmark scores**; strong Marathi rubric scoring; ~31B matches production scale
Primary (alt.)	Gemma-4-26B-A4B-it	26B MoE (3.8B active)	Apache 2.0	IndicParam 42.55%; near-31B quality at ~4B active inference cost — MoE fallback if dense 31B VRAM is tight
Cross-judge	Qwen2.5-72B-Instruct	72B	Apache 2.0	Different model family from Gemma; larger than production; strong JSON output — independent ensemble signal
Cross-judge (alt.)	Meta-Llama-3.3-70B-Instruct	70B	Llama license	Third family option if Gemma + Qwen correlate too highly on calibration set
Rubric specialist (optional)	prometheus-2-8x7b-v2.0	8×7B MoE	Apache 2.0	Purpose-built LLM-as-judge; supplementary third opinion on dimension scores only
Roman Marathi auxiliary	RomanSetu SFT native	7B	Apache 2.0	AI4Bharat Roman-script Indic; run **only** on `roman_input_handling` when query is Roman Marathi
Dispute / deep review	Qwen3-235B-A22B	235B MoE (22B active)	Apache 2.0	Tie-breaker when Gemma-4-31B and Qwen2.5-72B disagree by ≥2 points; monthly deep review of exemplars

Recommended default ensemble:

```
Primary:   Gemma-4-31B-it          # best open Indic scores; Marathi rubric anchor
Cross:     Qwen2.5-72B-Instruct    # larger, independent family
Dispute:   Qwen3-235B-A22B         # ≤10% of K
Auxiliary: RomanSetu-7B            # Roman-script queries only
```

Gemma 4 judge configuration:

- **Disable thinking mode** for weekly batch scoring — omit `<|think|>` from system prompt; faster, reproducible JSON dimension scores
- **Enable thinking mode** only on dispute-tier traces where judges disagree
- Use `temperature=0` for scoring (overrides Gemma 4's default `temperature=1.0` recommended for creative generation)

Not recommended: English-only judges as sole scorer; deprecated **Gemma 3** variants (superseded on Indic benchmarks by Gemma 4); any single model without ensemble.

Benchmarking (model selection, not production): IndicEval, IndicGenBench, IndicParam.

7.2 Infrastructure — weekly batch at 31B–72B scale

Concern	Guidance
VRAM	Gemma-4-31B ≈ 62 GB BF16; Qwen2.5-72B ≈ 144 GB BF16; tensor parallelism on dedicated eval GPUs
Throughput	K=500 × 2 judges ≈ 1,000 inferences/week — overnight run; Monday scorecard
Isolation	Eval GPUs separate from production 30B serving
Quantisation	4-bit OK for cross-judge (Qwen 72B); **keep Gemma-4-31B primary at BF16** for calibration stability
MoE fallback	Gemma-4-26B-A4B-it if dense 31B does not fit eval GPU pool

7.3 Bhili — recommended judge stack

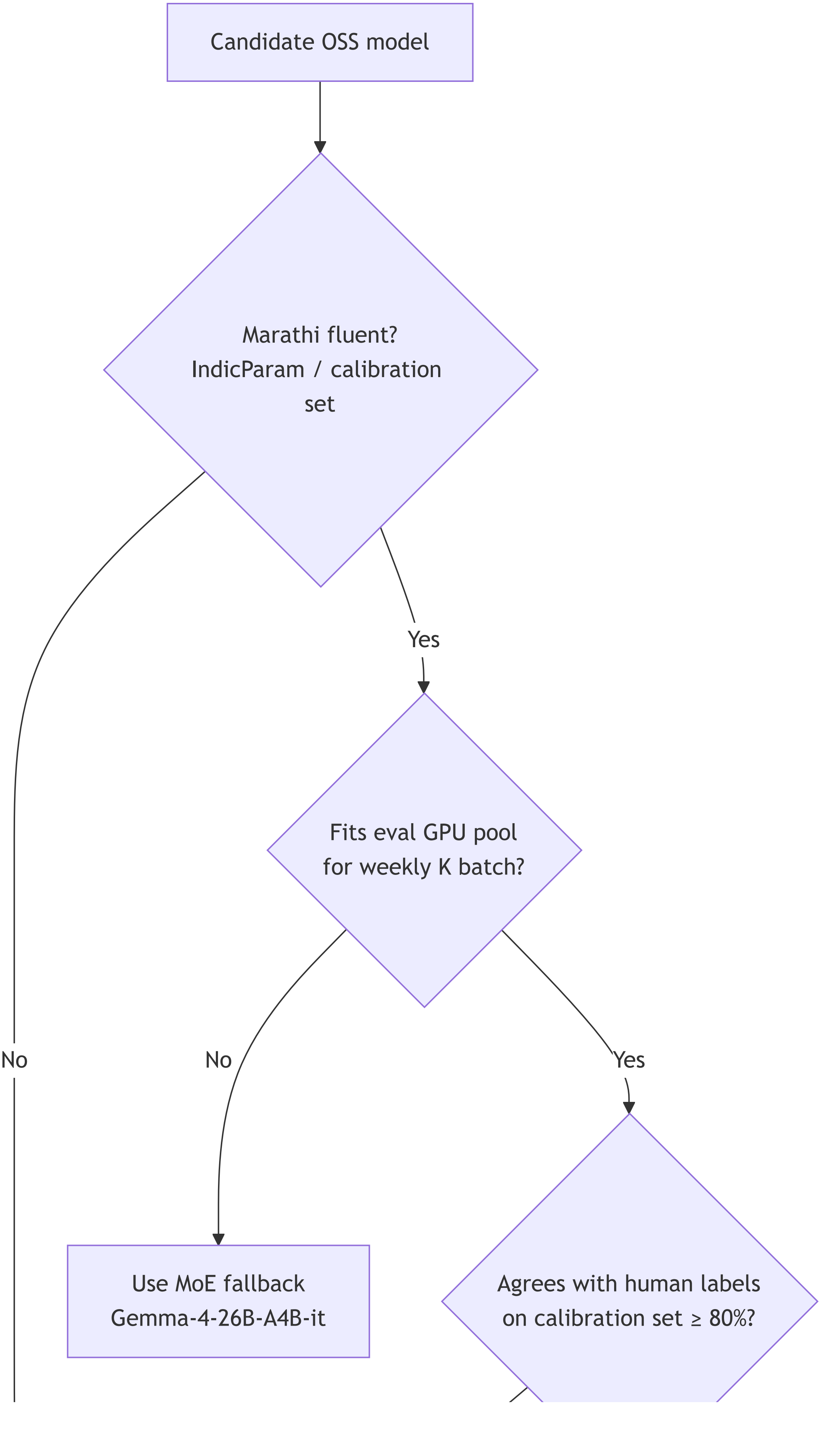
No mature open-source Bhili LLM-as-judge exists today. Do **not** rely on small MT adapters as judges.

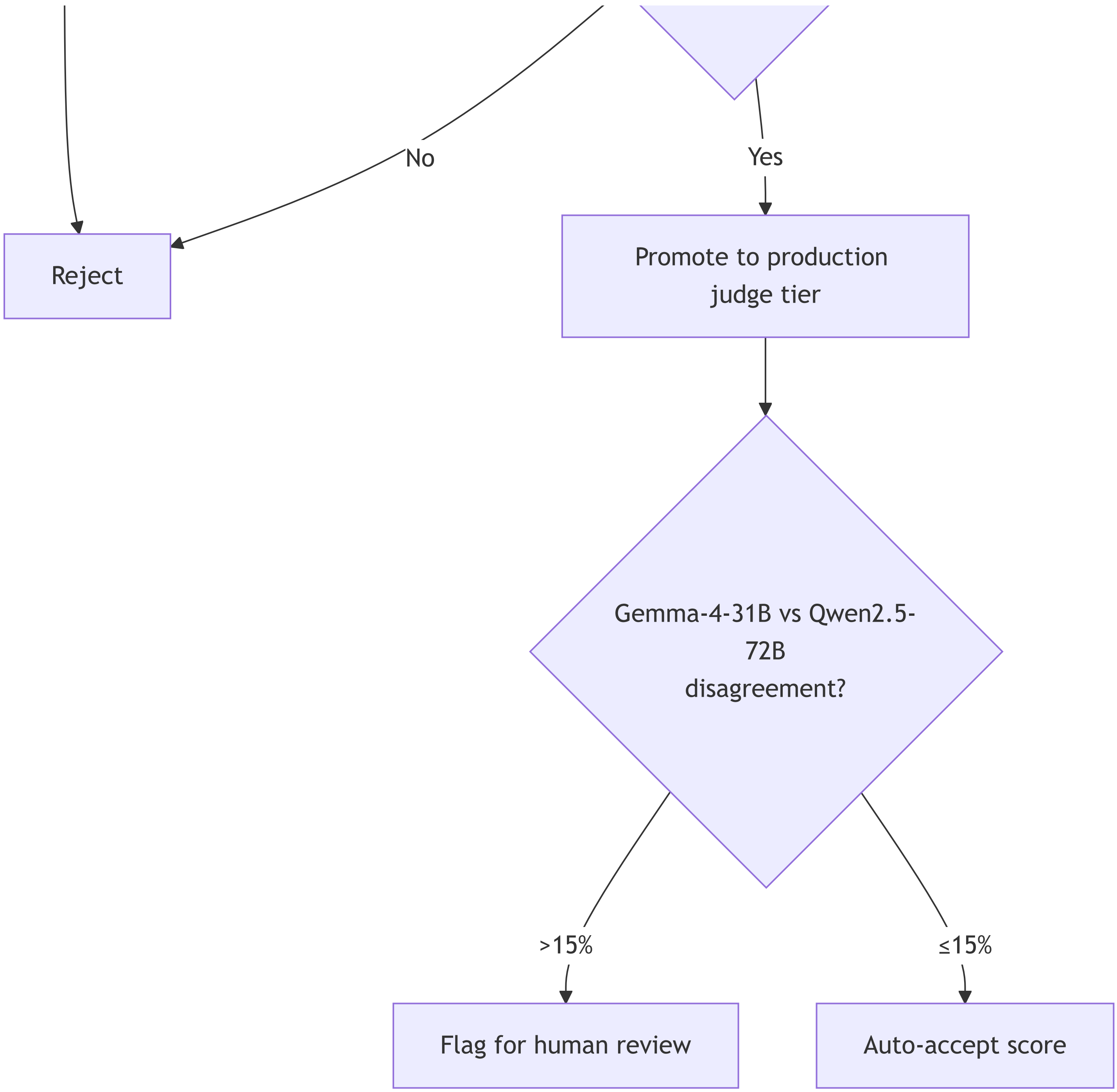
Asset	Source	Role in evaluation
Bhili glossary (`mar2bhb` term pairs)	AI4Bharat / field team	**Deterministic** crop/scheme name validation — primary Bhili check in P1
AdiVaani parallel corpus	IIIT-H / community	Fine-tuning data for Bhili judge LoRA on **Gemma-4-31B** base
Bhili ASR/TTS collection	AI4Bharat	Future voice-loop evaluation
Maharashtra Bhili Tribal LLM	State initiative (AI4Agri2026)	Target production + judge when weights are on-prem
Gemma-4-31B-it (§7.1)	Google DeepMind	Semantic review of Bhili responses against Marathi source advisory

Interim Bhili recipe:

1. **Rule layer** — `mar2bhb` glossary: flag Marathi terms left in Bhili output
2. **Large-model semantic check** — Gemma-4-31B-it compares Bhili response vs Marathi source (meaning, register, glossary)
3. **Human review** — 100% of Bhili pilot traces weekly
4. **Future** — LoRA judge adapter on Gemma-4-31B-it using AdiVaani + agri parallel data

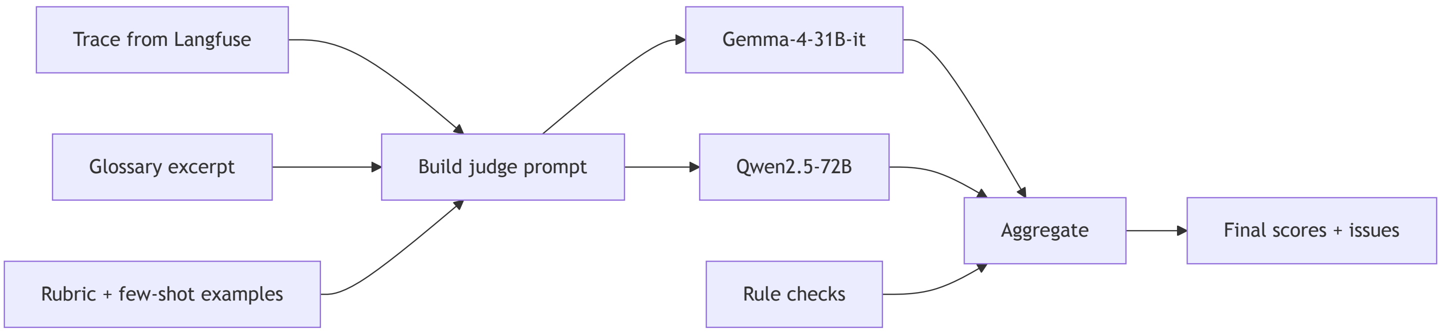
7.4 Model selection decision matrix


Candidate OSS model
Marathi fluent?
IndicParam / calibration
set
Yes
Fits eval GPU pool
for weekly K batch?
No
No
Yes
Use MoE fallback
Gemma-4-26B-A4B-it
Agrees with human labels
on calibration set ≥ 80%?


No
Yes
Reject
Promote to production judge tier
Gemma-4-31B vs Qwen2.5-72B disagreement?
>15%
≤15%
Flag for human review
Auto-accept score

8. How LLM-as-judge works (detailed)

8.1 Pipeline per conversation



Prompt contents (Marathi sessions):

1. System: role as Marathi agricultural advisory quality reviewer
2. Rubric: dimension definitions with 1–5 scale anchors **written in Marathi**
3. Context: user query, assistant response, `source_lang` , `target_lang` , moderation outcome
4. Glossary excerpt: relevant term pairs from authoritative glossary
5. Policy excerpts: language adherence rules (no English in Marathi replies, farmer-friendly register)
6. Output schema: JSON with scores, typed issues, evidence quotes, suggestions

Critical design choices:

Practice	Rationale
Rubric in **target language**	Reduces English-centric bias in Marathi scoring
Two judge families (Gemma-4-31B + Qwen2.5-72B)	Cuts single-model blind spots; disagreement → human queue
Different model than production agent	Avoids self-enhancement bias
Structured JSON output	Parseable; feeds Langfuse scores API
Evidence quotes	Reviewers verify without re-reading full trace
Temperature 0	Reproducible weekly scores
Gemma 4 thinking off for batch	Faster scoring; thinking on for dispute tier only

8.2 Rule layer (non-LLM)

Run **before** LLM judges — fast, deterministic:

- Regex / dictionary scan for Latin-script words in Marathi-target responses
- Match agricultural terms against glossary entries for the query's domain
- Detect selected-language mismatch (script heuristics)
- For Bhili: apply `mar2bhb` glossary substitution audit

Rule violations are **merged** with LLM issues; rules win on factual glossary conflicts.

8.3 Ensemble aggregation

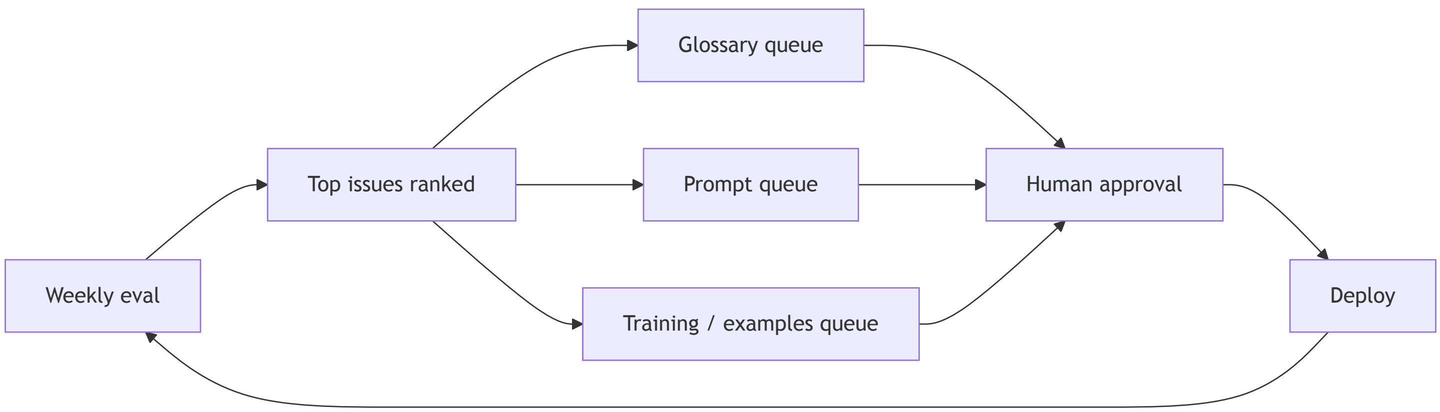
Signal	Weight
Gemma-4-31B score	40%
Qwen2.5-72B score	40%
Rule layer pass/fail	20% (binary penalty on affected dimensions)

If `|score_A − score_B| ≥ 2` on any dimension → trace goes to **human review queue**; optionally invoke Qwen3-235B-A22B dispute tier.

8.4 Human calibration (monthly)

- 5% of weekly K rated by Marathi-speaking agri extension reviewers (+ Bhili native speakers for tribal traces)
- Compute Pearson correlation and Cohen's κ between human and ensemble
- **Gate:** if language_quality κ < 0.6, pause scorecard publication and recalibrate rubric

9. Strategy to improve language quality



9.1 Issue → action mapping

Issue type	Typical action	Owner
`english_leakage`	Add/fix glossary entry; add negative example to system prompt	Eng + dept terminology
`glossary_mismatch`	Correct glossary term; audit `search_terms` matches	Eng
`register_too_formal`	Add farmer-friendly exemplar to prompt	Eng
`wrong_term_from_roman_input`	Improve Roman transliteration handling in `search_terms` examples	Eng
`language_adherence_failure`	Strengthen language adherence section; check `target_lang` injection	Eng
`bhili_marathi_residual`	Extend Bhili glossary; tune post-edit rules	Eng + tribal language team
`systematic_tool_terminology`	Fix upstream document/search content or term mapping	Eng + content

9.2 Suggestion types the judge produces

1. **Inline language correction** — "Replace X with Y" for a specific trace
2. **Glossary addition** — new `en → mr` (or `mr → bhb`) pair proposal
3. **Prompt gap** — missing guidance in system prompt → engineering ticket

4. **Positive exemplar** — high-scoring traces as few-shot candidates (monthly curation)

9.3 Prioritisation

Rank issues by: `frequency × severity × farmer–facing impact`. Weekly scorecard leads with **top 5**. Department meeting: approve glossary/prompt changes for top 3.

9.4 Bhili-specific improvement path

1. **Pilot** — 100% human review; LLM + glossary assist
2. **Collect failures** — Marathi leakage, dialect violations
3. **Fine-tune** — LoRA judge adapter on Gemma-4-31B-it using AdiVaani + agri parallel sentences
4. **Integrate tribal LLM** — when Maharashtra Bhili Tribal LLM is on-prem, swap into judge tier with continued human calibration

10. Judge dimensions (full scorecard)

Area	Weight	Notes
Language quality	High	Primary focus — see §6
Agricultural relevance	High	Addresses the farmer's actual question
Terminology & glossary	High	Tied to language quality
Safety & moderation	Medium	Consistent with moderation outcome
Tool / MCP use	Medium	Sensible calls; failures annotated
Reliability	Medium	Non-empty response; errors in trace
Latency	Low	Informational; not a language issue

11. Weekly quality scorecard

Section	Contents
Header	Week ID, date range, K reviewed, environments
Language summary	Mean `language_quality` by `target_lang`; WoW delta
Top issues	Counts by `issue.type`; top 5 with exemplar trace IDs
Glossary queue	New term proposals from judge
Prompt queue	`prompt_gap_note` items
Disputed traces	Judge disagreement or human-queue backlog
Reliability	Error rate, empty responses, tool failures
Trend	4-week sparkline for language and safety
Actions taken	Changes deployed since last week (manual log)

12. Configuration

Parameter	Purpose	Example
`EVAL_K`	Conversations per weekly batch	`500`
`EVAL_WINDOW_DAYS`	Lookback window	`7`
`EVAL_SCHEDULE`	Cron expression	`0 6 * * 1` (Mon 06:00 IST)
`EVAL_ENV`	Trace filter	`production`
`JUDGE_MODEL_PRIMARY`	Main Marathi/Indic judge	`gemma-4-31b-it`
`JUDGE_MODEL_CROSS`	Second judge family	`qwen2.5-72b-instruct`
`JUDGE_MODEL_PRIMARY_ALT`	MoE fallback	`gemma-4-26b-a4b-it`
`JUDGE_MODEL_DISPUTE`	Tie-breaker	`qwen3-235b-a22b`

Parameter	Purpose	Example
`JUDGE_MODEL_ROMAN`	Roman-script auxiliary	`romansetu-base-sft-native`
`JUDGE_BHILI_ENABLED`	Phase gate	`false` until Bhili live
`JUDGE_GEMMA_THINKING`	Thinking mode	`false` for batch; `true` for dispute tier
`LANGFUSE_*`	Read traces, write scores	—
`GLOSSARY_PATH`	Authoritative term file	bundled glossary JSON
`HUMAN_CALIBRATION_RATE`	Monthly calibration fraction	`0.05`

13. Implementation status

Item	Status
Logfire / Pydantic AI `instrument=True`	In production repo
Langfuse export on chat path	**Not implemented** — blocking milestone
Weekly eval worker	**Not implemented**
OSS judge deployment	**Not implemented**
Glossary rule layer	**Not implemented**
Vistaar `OE_*` telemetry	Partial; separate from this framework

First milestone: Langfuse receives complete chat traces for every production conversation.

Second milestone: Weekly job on staging with K=50; human calibration before department-facing scorecard.

14. Minimum trace fields

- `session_id` , `source_lang` , `target_lang`
- Full user query and complete assistant response (not truncated)
- Moderation classification and outcome
- Tool / MCP invocations with names, inputs, errors
- Timestamp, environment, trace ID

15. Risks and mitigations

Risk	Mitigation
LLM judge wrong on Marathi nuance	Gemma-4-31B + Qwen2.5-72B ensemble + monthly human κ gate
Bhili judging unreliable	Phase human-first; glossary rules; no auto-actions
Eval GPU cost at 31B–72B scale	Weekly batch only; eval GPUs isolated from production; dispute tier ≤10% of K
PII in traces	Mask before judge prompt; same policy as production Agristack handling
Scorecard fatigue	Top-5 issues only; actionable queues; WoW trend not raw dumps
Prompt injection via farmer query	Judge sees moderation outcome; sandboxed JSON parsing

16. Open questions

1. Confirm Langfuse project and retention policy for 7-day lookback
2. GPU allocation for Gemma-4-31B-it + Qwen2.5-72B weekly batch (separate from 30B production serving)
3. Department reviewer roster for monthly Marathi calibration
4. Bhili tribal language team availability for pilot human review

5. Whether Maharashtra Bhili Tribal LLM weights will be available for on-prem judge tier
6. Integration of scorecard into existing Vistaar department dashboard vs Langfuse-only

17. References

- Doğruöz et al. (2026). *Challenges and Recommendations for LLMs-as-a-Judge in Multilingual Settings and Low-Resource Languages.* arXiv:2607.02235
- Google DeepMind. *Gemma 4* model card — ai.google.dev/gemma/docs/core/model_card_4
- IndicParam benchmark — arXiv:2512.00333; Gemma-4-31B-it: 46.48% overall
- AI4Bharat. *Bhili on MahaVISTAAR* collection; *RomanSetu* for Roman-script Indic
- Qwen Team. *Qwen2.5* technical report — arXiv:2407.10671
- Kim et al. (2024). *Prometheus 2* — open-source rubric judge. arXiv:2405.01535
- IndicEval — Indic LLM evaluation suite (model selection benchmarking)
- Langfuse docs — traces, sessions, scores API

Weekly loop: sample K conversations from Langfuse → Gemma-4-31B-it + Qwen2.5-72B judges on our infra → language-focused scores and correction suggestions → Langfuse + scorecard → human-approved glossary and prompt improvements.